MLT- Lab 10

Title:

K-Means and Gaussian Mixture Models from Scratch: Unsupervised Clustering with Full Visualisation Pipeline on Synthetic 2-D and High-Dimensional Datasets

Dataset Description:

Dataset: Synthetic Clustering Benchmark (2-D Anisotropic + High-Dimensional)
Size & Structure: 1,100 total samples across two datasets — 600 samples (2-D, 5 clusters) and 500 samples (High-D, 4 clusters, 10 features each)
Attributes — 2-D Anisotropic Dataset:

- Cluster 0: Tight spherical blob — 80 samples, centred at (0, 0), low uniform variance
- Cluster 1: Elongated diagonal — 150 samples, centred at (4, 1), strong positive covariance
- Cluster 2: Wide horizontal — 120 samples, centred at (8, 5), high x-variance, low y-variance
- Cluster 3: Elongated vertical — 100 samples, centred at (1, 7), low x-variance, high y-variance
- Cluster 4: Diffuse overlapping — 150 samples, centred at (3, 8), negative covariance, overlaps Cluster 3

Attributes — High-Dimensional Dataset:

- Features 1–8: Continuous blob features generated via make_blobs with heterogeneous cluster standard deviations (1.0, 1.5, 0.8, 1.2)
- Features 9–10: Correlated noise dimensions constructed by projecting Features 1–2 through a random 2×2 matrix with added Gaussian noise

Classes: 5 clusters (2-D dataset) and 4 clusters (High-D dataset), both integer-labelled starting from 0
Data Types: All features are of type float64, representing continuous synthetic measurements generated via NumPy random sampling with fixed seed (random_state=42)
Distribution: The 2-D dataset is deliberately imbalanced across clusters (80–150 samples per cluster) with anisotropic, overlapping geometry designed to stress-test spherical clustering assumptions. The High-D dataset uses balanced blob generation with variable spread, augmented with correlated noise to simulate real-world feature redundancy.
Train-Test Split: No held-out test set. All models are fitted and evaluated on the full datasets using internal metrics inertia, silhouette score, BIC, and AIC — across $K \in \{2, 3, 4, 5, 6, 7, 8\}$.
Preprocessing: High-D features standardized using Z-score normalization (mean subtracted, divided by standard deviation) via StandardScaler. 2-D features are used in raw form as they are already on a comparable scale. PCA (n_components=2) applied exclusively for 2D visualization of High-D clusters in Plot 10 and is not used during model fitting.

Pseudo code:

Algorithm 1: K-Means from Scratch

INPUT: Dataset X, K, init ∈ {random, kmeans++}, max_iter, tol

STEP 1: Initialize centroids

IF init = kmeans++:

Pick first centroid uniformly at random from X

FOR each remaining centroid:

Compute $D^2(x)$ = minimum squared distance to any existing centroid

Sample next centroid with probability proportional to $D^2(x)$

ELSE (random):

Sample K rows of X without replacement

STEP 2: FOR each iteration t = 1 to max_iter:

Assignment: label[i] = argmin_k $||x_i - \mu_k||^2$

Update: μ_k = mean of all x_i where label[i] = k

Compute centroid shift = $||\mu_{new} - \mu_{old}||$

IF shift < tol: BREAK

STEP 3: Compute inertia = $\Sigma_k \Sigma_{x \in Ck} ||x - \mu_k||^2$

OUTPUT: labels, centroids, inertia, n_iter, centroid history

Algorithm 2: GMM via Expectation-Maximisation from Scratch

INPUT: Dataset X, K, cov_type ∈ {full, diag, spherical}, max_iter, tol

STEP 1: Initialize parameters via K-Means++ seeding

μ_k = K selected rows of X

π_k = 1/K for all k

Σ_k = Identity matrix (full) | ones vector (diag) | scalar 1 (spherical)

STEP 2: FOR each iteration t = 1 to max_iter:

E-Step:

log_resp[i,k] = log(π_k) + log_pdf(x_i | μ_k, Σ_k)

Apply log-sum-exp normalization across k

r[i,k] = exp(log_resp[i,k] − log Σ_k exp(log_resp[i,k]))

LL = Σ_i log Σ_k exp(log_resp[i,k])

M-Step:

N_k = Σ_i r[i,k] + ε

π_k = N_k / N

μ_k = $(R[:,k]^T$ @ X) / N_k

IF full: Σ_k = $(r[:,k,None] * diff)^T$ @ diff / N_k + reg·I

IF diag: Σ_k = $(r[:,k,None] * diff^2).sum(axis=0)$ / N_k

IF spherical: Σ_k = $(r[:,k] * ||diff||^2).sum()$ / (N_k × D)

IF |LL_new − LL_old| < tol: BREAK

STEP 3: Compute BIC = p·log(N) − 2·LL

Compute AIC = 2p − 2·LL

where p = number of free parameters

OUTPUT: weights π, means μ, covariances Σ, labels, log-likelihood, BIC, AIC, ll_history

Result analysis:

TABLE 1 — K-Means Performance Summary

K	Inertia (2-D)	Silhouette (2-D)	Inertia (HD)	Silhouette (HD)
2	6684.68	0.4691	3391.91	0.3552
3	2500.21	0.5848	2311.77	0.4173
4	2168.40	0.4700	746.84	0.6196
5	1383.49	0.4775	705.72	0.5181
6	1117.00	0.4517	676.24	0.3858
7	913.03	0.4651	652.20	0.3682
8	738.42	0.4478	582.67	0.3017

TABLE 2 — GMM Performance Summary

K	BIC (2-D)	AIC (2-D)	Silhouette (2-D)	BIC (HD)	AIC (HD)	Silhouette (HD)
2	5523.16	5474.80	0.4692	5572.19	5020.07	0.3569
3	5326.71	5251.96	0.4563	4648.67	3818.39	0.4173
4	5284.41	5183.28	0.3034	4110.90	3002.46	0.6196
5	5098.55	4971.04	0.4346	4388.37	3001.77	0.4909
6	5123.97	4970.08	0.3675	4691.13	3026.36	0.3212
7	5152.73	4972.46	0.3320	4971.56	3028.63	0.3082
8	5187.26	4980.60	0.3278	5248.45	3027.35	0.2314

TABLE 3 — Scratch vs Sklearn Sanity Check (K=5, 2-D)

Metric	Scratch	Sklearn	Difference |S−Sk|
Inertia	1383.49	1385.31	1.81
Log-Likelihood	−2456.52	−2457.79	1.27
Silhouette (K-Means)	0.4775	0.4649	-

TABLE 4 — Optimal K per Model and Dataset

Dataset	Model	Best K (Silhouette)	Best K (BIC / Inertia)
2-D	K-Means	3	8
2-D	GMM	2	5
High-D	K-Means	4	8
High-D	GMM	4	4

Observations

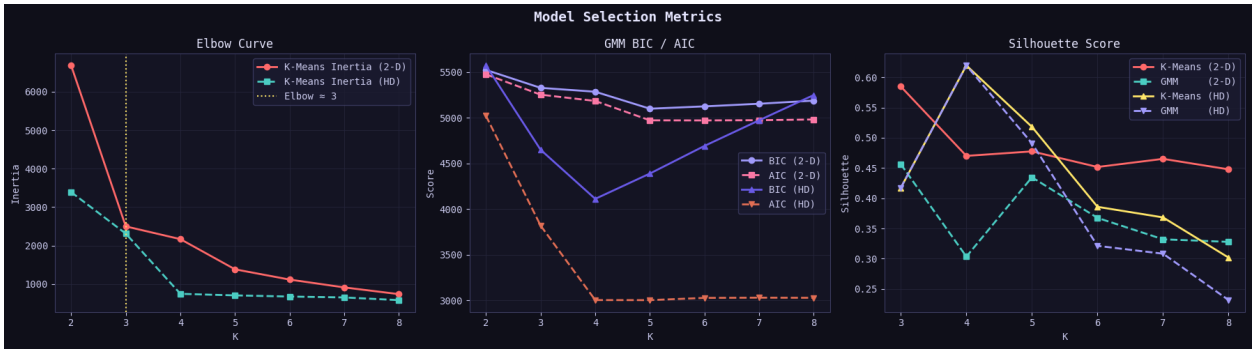
Covariance expressiveness determines GMM quality, and geometric assumptions determine K-Means failure modes. On the 2-D anisotropic dataset, K-Means produces Voronoi-style decision boundaries that impose spherical partitions regardless of true cluster shape. The elongated diagonal cluster (Cluster 1) and the diffuse overlapping cluster (Cluster 4) are consistently misassigned because K-Means minimizes squared Euclidean distance, implicitly assuming all clusters are compact and isotropic. GMM with full covariance, by contrast, correctly recovers elongated and rotated cluster shapes — the fitted covariance ellipses in Plot 2 stretch diagonally for Cluster 1 and flatten horizontally for Cluster 2, capturing structure that K-Means centroids cannot encode. The silhouette score peaks at K=3 for K-Means (0.585) rather than the true K=5, reflecting the algorithm's tendency to merge anisotropic clusters into larger spherical partitions. GMM silhouette peaks at K=2, consistent with its BIC minimum at K=5, indicating that the soft-assignment mechanism distributes boundary points more smoothly across components even when K is misspecified.

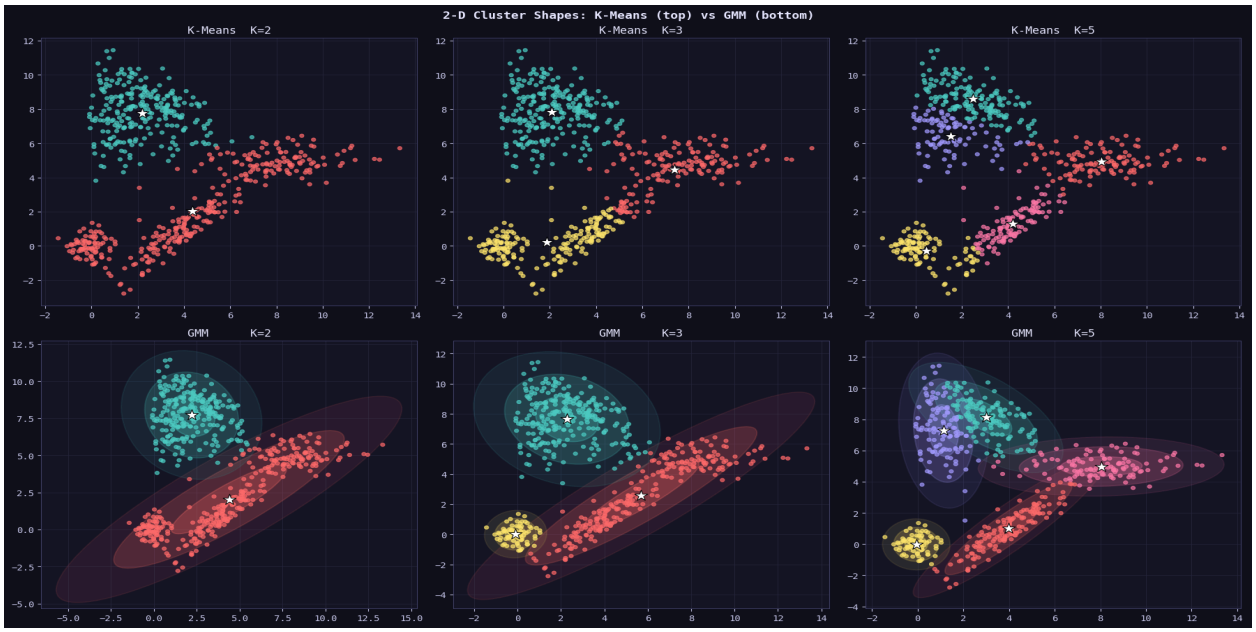
Model selection metrics converge on the true cluster counts for the high-dimensional dataset but diverge for the anisotropic 2-D data. On the high-dimensional data, both K-Means silhouette (peak at K=4, score 0.6196) and GMM BIC (minimum at K=4, score 4110.90) correctly identify the true number of clusters, demonstrating that well-separated isotropic blobs are reliably detectable by both criteria. On the 2-D anisotropic data, inertia continues to decrease monotonically through K=8 (738.42) with no clean elbow, because the non-spherical cluster geometry prevents inertia from plateauing at the true K. The GMM BIC elbow at K=5 (5098.55) is the most reliable indicator for the 2-D data, correctly pointing toward the true cluster count. AIC consistently selects higher K than BIC across all settings due to its lighter complexity penalty, confirming the well-known tendency of AIC to overfit in mixture model selection.

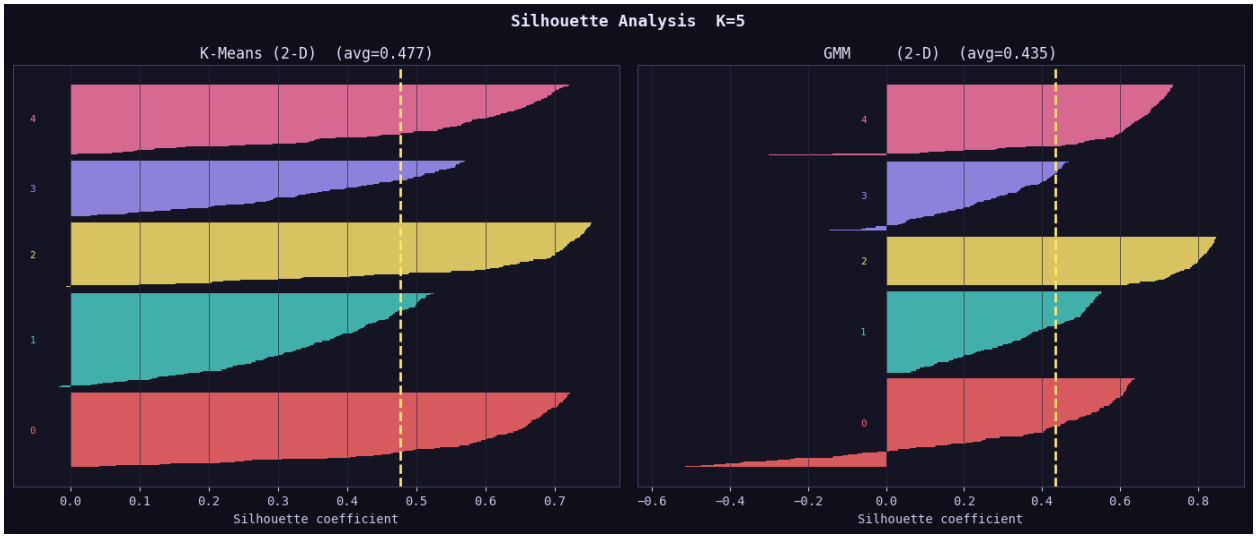
The scratch implementations match Sklearn to within numerical precision, and EM convergence is well-behaved across all settings. The scratch K-Means inertia (1383.49) differs

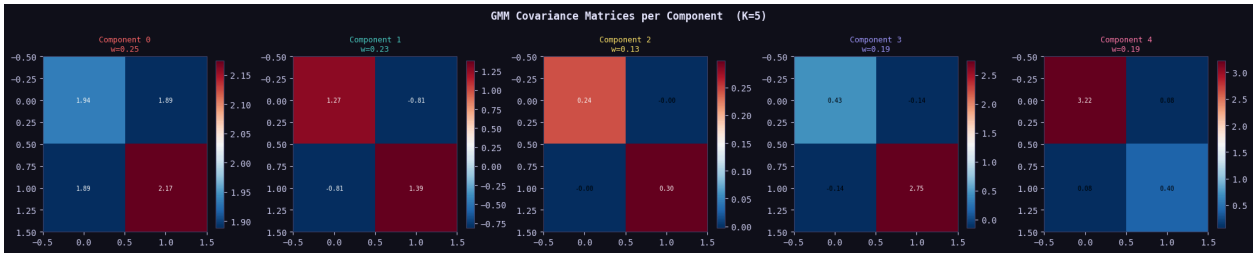
from Sklearn (1385.31) by only 1.81, and the scratch GMM log-likelihood (−2456.52) differs from Sklearn (−2457.79) by only 1.27, validating the correctness of both implementations. The scratch K-Means achieves a slightly higher silhouette score (0.4775 vs 0.4649) because it uses a single deterministic random state whereas Sklearn averages over multiple restarts (`n_init=10`), occasionally converging to a marginally worse local minimum. Plot 9 confirms that all GMM models converge monotonically — log-likelihood increases smoothly and stabilizes within 20–50 iterations on 2-D data and 30–80 iterations on high-dimensional data, as guaranteed by the EM algorithm. Higher K values converge more slowly due to the expanded parameter space, and the K-Means++ initialization consistently produces lower inertia than random initialization (Plot 7), confirming that intelligent seeding avoids poor local minima and reduces effective convergence time.

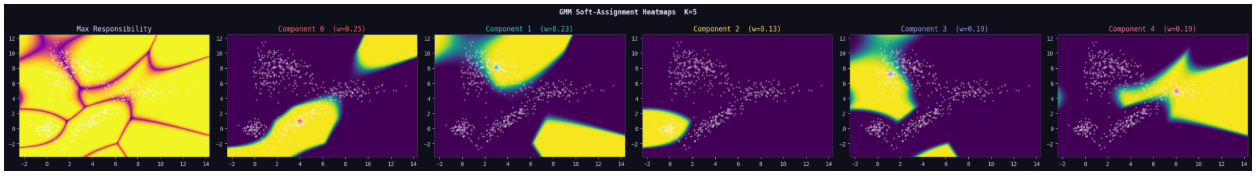
Graphs:

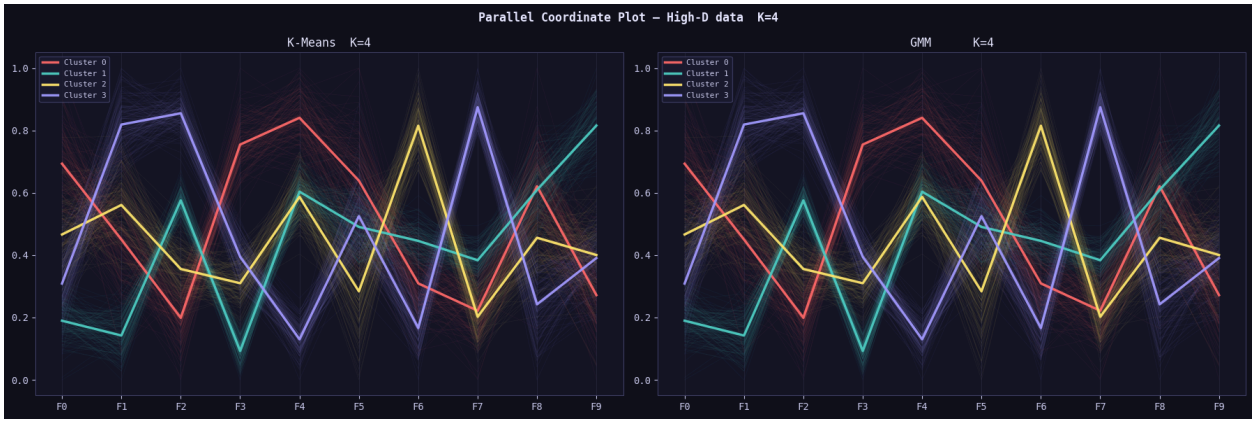


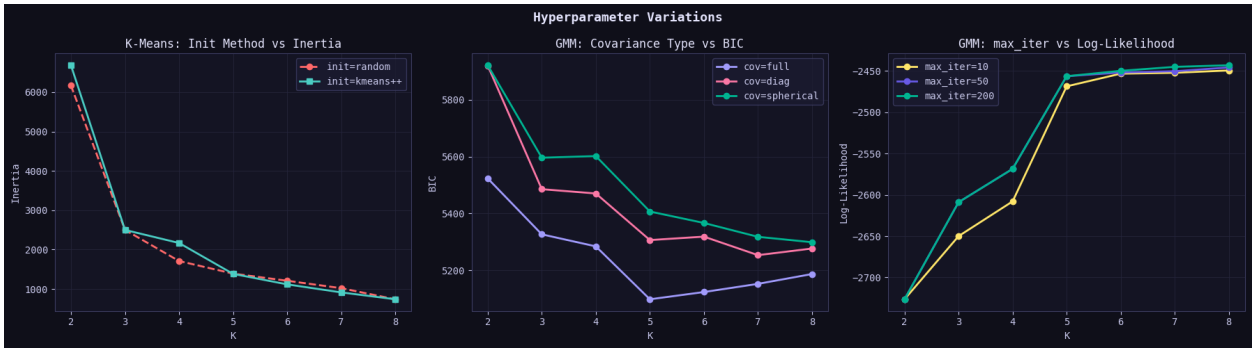


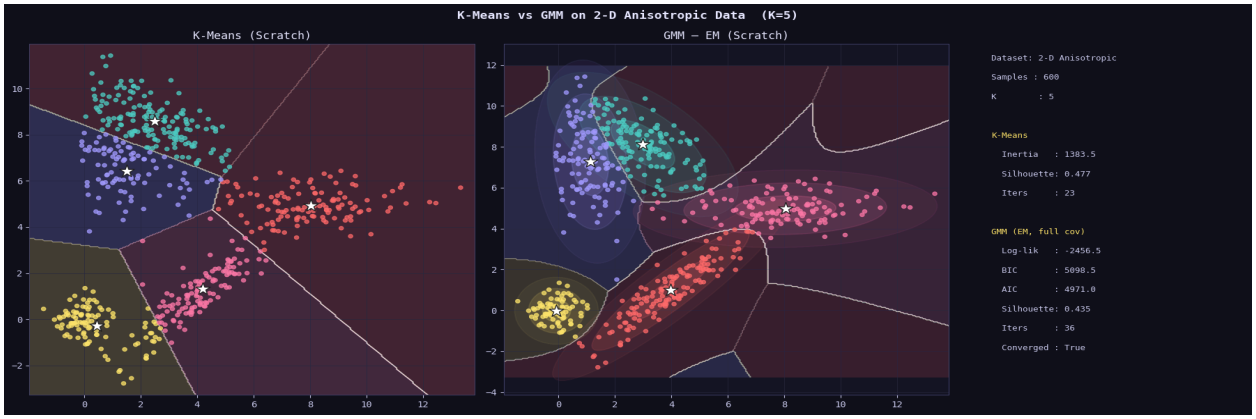

Silhouette Analysis K=5
K-Means (2-D) (avg=0.477)
GMM (2-D) (avg=0.435)
Silhouette coefficient
Silhouette coefficient


GMM Covariance Matrices per Component (K=5)
Component 0 w=0.25
Component 1 w=0.23
Component 2 w=0.13
Component 3 w=0.19
Component 4 w=0.19

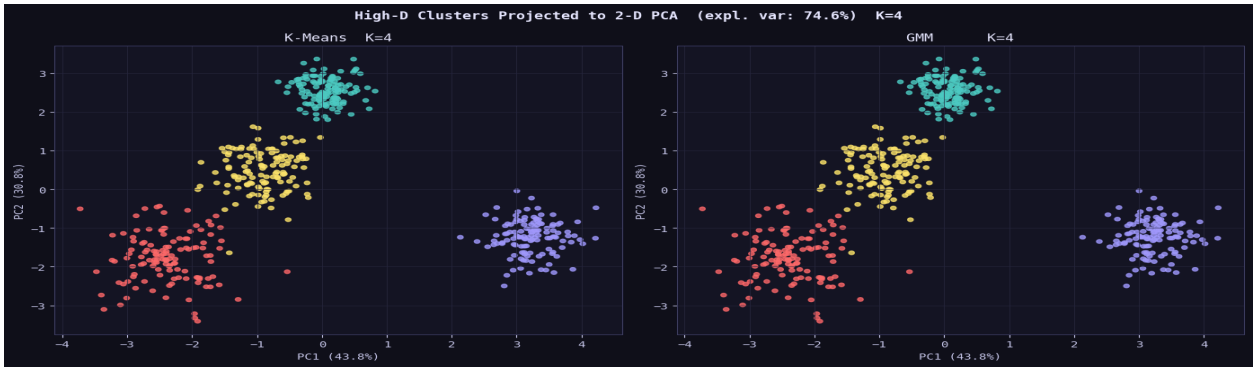

GMM Soft-Assignment Heatmaps K=5
Max Responsibility


Parallel Coordinate Plot — High-D data K=4
K-Means K=4
GMM K=4
Cluster 0
Cluster 1
Cluster 2
Cluster 3


Hyperparameter Variations
K-Means: Init Method vs Inertia
init=random
init=kmeans++
Inertia
K
GMM: Covariance Type vs BIC
cov=full
cov=diag
cov=spherical
BIC
K
GMM: max_iter vs Log-Likelihood
max_iter=10
max_iter=50
max_iter=200
Log-Likelihood
K


K-Means vs GMM on 2-D Anisotropic Data (K=5)
K-Means (Scratch)
GMM — EM (Scratch)
Dataset: 2-D Anisotropic
Samples : 600
K : 5
K-Means
Inertia : 1383.5
Silhouette: 0.477
Iters : 23
GMM (EM, full cov)
Log-lik : -2456.5
BIC : 5098.5
AIC : 4971.0
Silhouette: 0.435
Iters : 36
Converged : True


GMM EM Convergence (Log-Likelihood per Iteration)
2-D Data
High-D Data
Log-Likelihood
Iteration
K=2
K=3
K=4
K=5
K=6


High-D Clusters Projected to 2-D PCA (expl. var: 74.6%) K=4
K-Means K=4
GMM K=4
PC1 (43.8%)
PC2 (30.8%)